

The Winter Bottleneck: | Analyzing Dock Capacity Pressure in Boston's BLUEbikes

1

Context

- Urban bike systems rely on **balanced dock availability**
- **Winter weather** introduces demand shocks and operational instability

2

Problem

- Stations become **overfull** (no docks) or **empty** (no bikes)
- Leads to **service failure** and **reduce operational efficiency** despite system availability

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Objective

- Analyze **dock pressure** across stations
- Identify **what drives imbalance** by evaluating time, location and weather variables

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Key Metric Net Flow Pressure Index

Measures **real-time station stress**
Formula: $(Arrivals - Departures) / Dock Capacity$

Interpretation:
Positive = filling up (overflow risk)
Negative = emptying (shortage risk)

In code: Use `|Pressure Index|`
Focus: Magnitude of imbalance, not direction. Either way it's service failure

Data Source

#	Name/Description	Source URL	Type	Fields (Key Info)	Format	Data Size
1	Bluebikes Trip History (Nov 2025)	https://s3.amazonaws.com/hubway-data/index.html	File Download from Web	ride_id, started_at, ended_at, start_station_id, end_station_id, lat/long	csv	344,842 trips (13 columns)
2	Bluebikes Station List	https://bluebikes.com/system-data	File Download from Web	Station_id, lat, long, total_docks	csv	597 stations (8 columns)
3	Open-meteo Weather Data (Hourly)	https://open-meteo.com/en/docs/historical-weather-api	API	time, temperature, precipitation, snowfall, windspeed, cloud cover	JSON	~744 hourly records (6 columns)

Identify Patterns

Time analysis

Two clear peak hours:

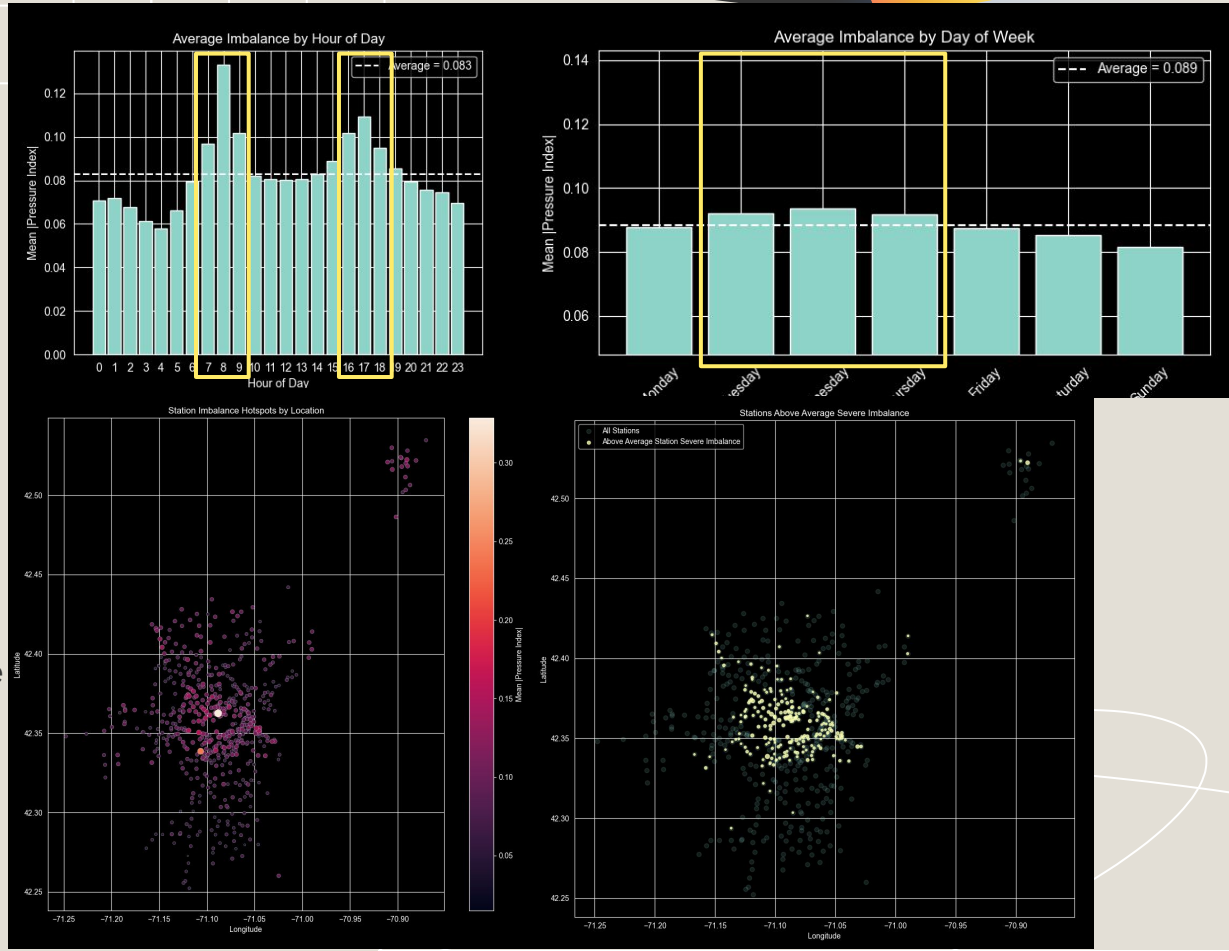
- Morning commute (7–9 AM)
- Evening commute (4–6 PM)

Three clear peak days:

- Tuesday - Thursday

Spatial analysis - Station-level Imbalance

- Imbalance is not evenly distributed
- ~38% stations above average imbalance
- ~49% stations above average severe imbalance ($|PI| > 0.2$)
- Hotspots cluster in central boston area



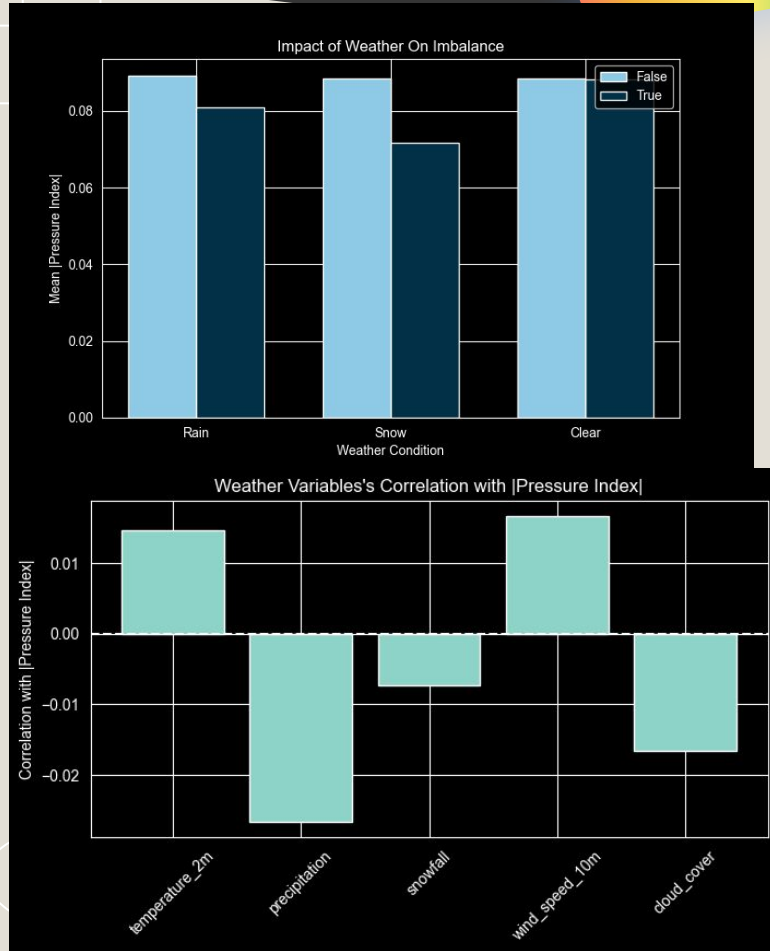
Weather Analysis

Initial Weather Analysis

- Rain & Snowfall associated with lower pressure imbalance level -> adverse weather does reduce system usage
- Snowfall showcase the strongest impact, with a gap of 0.016 when comparing if its is snowing or not.

Weather Correlation Analysis

- Very weak correlation overall
- Weather alone does not explain imbalance
- Rain -> slightly reduces imbalance
- Temperature -> slightly positive effect



Conditional Weather Analysis

Further explore the performance of weather impact during

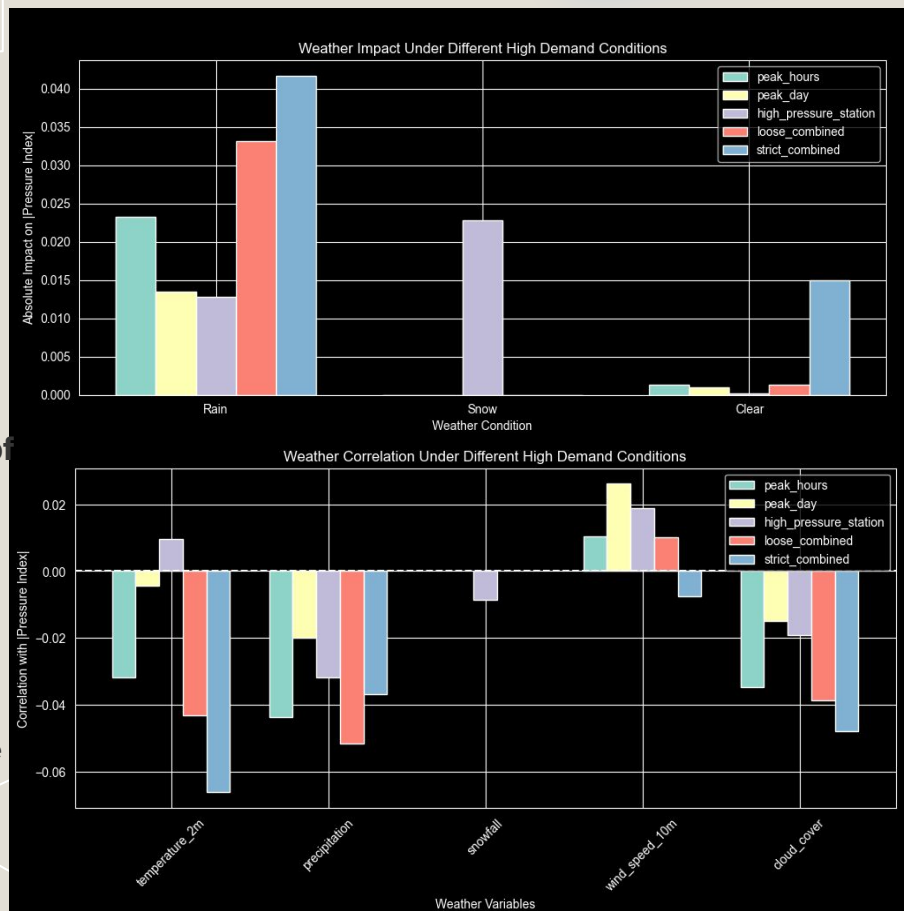
high demand conditions:

1. Peak Hours = Rush hour periods - 7,8,9,16,17,18
2. Peak Days = Weekday with strong demand - Tuesday, Wednesday, Thursday
3. High Pressure Station = station with above average mean pressure imbalance
4. Loose Combined = peak hours x high pressure stations
5. Strict Combined = peak hours x peak day x high pressure stations

The correlation value remain close to zero: **weather does not have a strong direct linear relationship with pressure imbalance.**

Weather impact is conditional on demand intensity: **an amplifier of existing system stress, not a primary drive**

1. **Temperature** shows the strongest relationship with imbalance
 - Lower temperature -> higher imbalance (negative correlation)
2. **Precipitation and cloud cover** consistently increase imbalance
 - Moderate negative correlations across most conditions
3. **Snowfall** shows inconsistent correlation
 - Near-zero or unstable relationship across conditions, suggests limited observable impact in current dataset, could also due to the small sample size.
4. **Wind speed** has minimal and inconsistent influence
 - Weak correlations overall, not a major driver of imbalance compared to other variables



Logistic Regression Model

Goal: Predict what leads to extreme imbalance, & does adding weather information help predict extreme pressure better than using only time and location related feature. (Extreme pressure is defined as $\text{abs_pressure_index} > 0.2$)

Models:

1. Baseline: Time + station feature
2. Weather: Time + station + weather

Weather Model Performance:

- Accuracy = 49.8% correctness, model is not very strong
- Precision = 11.6%, model produces many false positives
- Recall = 82.2%, model is good at finding extreme cases

Positive Coefficient -> Extreme Pressure:

- Highest feature - Location
- Second highest - Day of Week (Time)
- Third highest - Temperature (higher temp -> high demand -> imbalance)

Negative coefficient -> Less Extreme Pressure:

Lowest feature - precipitation (more rain -> low demand -> less imbalance)

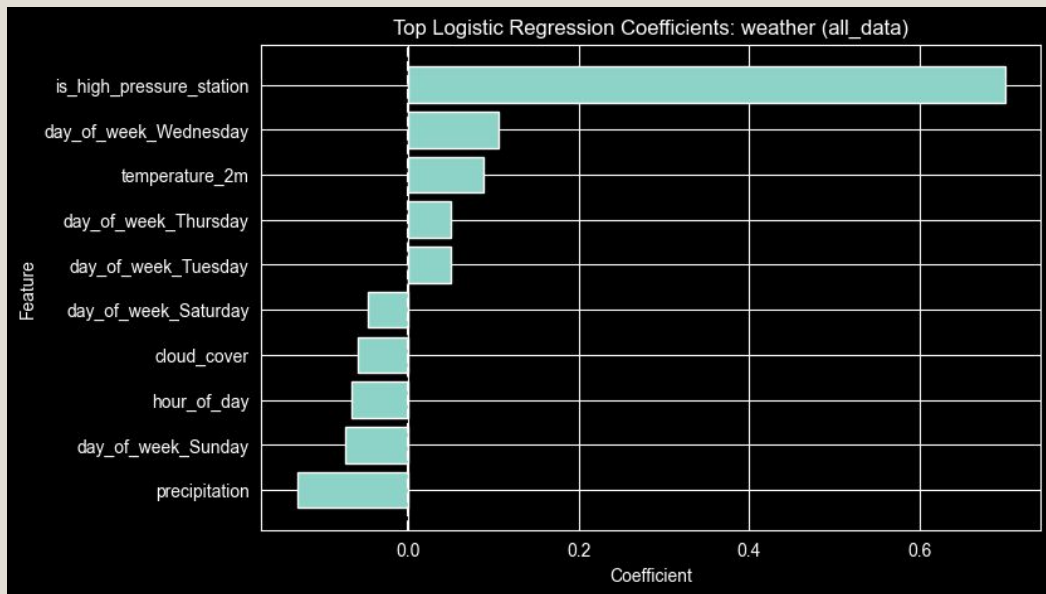
Model confirms findings: non-linear

Time + location explain most variation

Weather adds small improvement

Practical Implication:

Focus rebalancing on: Peak hours & Stations



Challenges

Challenge 1	Weather Data Resolution Constraints	Ideally, weather conditions should be matched precisely to each station using the coordinates (longitude and latitude). But, weather API only support either single coordinate or grid level weather data. Therefore, project only uses a centralized weather approximation
Challenge 2	Pressure Index Simplification	Pressure index is used as absolute value, ignoring the inflow and outflow of each station, thereby unable to identify the exact dock capacity. Bluebike operations include manual rebalancing that are not captured in the dataset, so it limit the ability to reflect actual dynamics. However, still useful to see system level imbalance patterns.
Challenge 3	Lack of Interaction Model	This model uses linear regression and logistic regression, but the system clearly involves interaction between weather, time, and location. Should consider adding random forest to present the non linear relationship.